

Protein Supply and Contribution by Major Food Groups in Bangladesh, 2010–2023

1. Introduction

This analysis examines changes in the protein supply of Bangladesh between 2010 and 2023 using six major food groups: **Fish, Meat, Milk, Eggs, Pulses, and Cereals**. The objective is to identify changes in the quantity of protein supplied by each food group, examine their relative contribution to total protein supply among the six selected groups, and compare the contribution of animal-source and plant-source foods.

The analysis uses protein supply values measured in **grams per capita per day (g/capita/day)**. Food items were classified into the six selected groups according to the food-group classification used for the analysis. Individual component items were summed within each group for each year. Aggregate categories were excluded where their component items were already included in order to avoid double counting. For Milk and Eggs, the main aggregate measures **Milk - Excluding Butter** and **Eggs** were retained.

2. Method of Analysis

The original food-item data were first classified into six food groups. The Fish group included freshwater fish, demersal fish, pelagic fish, marine fish and other aquatic products. The Meat group consisted of bovine meat, mutton and goat meat, poultry meat, and other meat. Pulses included beans, peas, and other pulses and products. Cereals included rice, wheat, barley, maize, rye, oats, millet, sorghum, and other cereals. Milk was represented by Milk - Excluding Butter, while Eggs was represented by the aggregate Eggs measure.

For each year, protein supply was calculated by summing the values of all included items belonging to the relevant food group.

The absolute change between 2010 and 2023 was calculated as:

$$\text{Absolute Change} = \text{Protein supply in 2023} - \text{Protein supply in 2010}$$

The percentage change was calculated as:

$$\text{Percentage Change} = [(\text{2023 value} - \text{2010 value}) / \text{2010 value}] \times 100$$

The contribution of each food group was calculated as its protein supply divided by the combined protein supply of the six selected food groups in the same year.

For the animal–plant comparison, Fish, Meat, Milk, and Eggs were classified as **animal-source foods**, while Pulses and Cereals were classified as **plant-source foods**.

3. Protein Supply Trend

The results show that protein supply increased between 2010 and 2023 for five of the six selected food groups, while cereal-derived protein declined slightly.

Food group	2010	2023	Absolute change	Percentage change
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Fish	5.64	7.90	+2.26	+40.07%
Meat	1.78	2.05	+0.27	+15.17%
Milk	2.00	6.95	+4.95	+247.50%
Eggs	0.46	1.18	+0.72	+156.52%
Pulses	3.22	5.66	+2.44	+75.78%
Cereals	36.90	35.78	−1.12	−3.04%

Among the six food groups, **milk experienced the largest increase in protein supply**, rising from 2.00 g/capita/day in 2010 to 6.95 g/capita/day in 2023. This represents an increase of 4.95 g/capita/day, or **247.50%**.

Egg protein supply also increased substantially, from 0.46 to 1.18 g/capita/day, representing a **156.52% increase**. Pulses increased from 3.22 to 5.66 g/capita/day, equivalent to a **75.78% increase**, while fish increased from 5.64 to 7.90 g/capita/day, or **40.07%**.

Meat showed a comparatively smaller increase, rising from 1.78 to 2.05 g/capita/day, representing a **15.17% increase**.

In contrast, cereal-derived protein declined slightly from 36.90 to 35.78 g/capita/day, a reduction of **1.12 g/capita/day or 3.04%**. Despite this decline, cereals remained by far the largest contributor to protein supply among the six selected food groups.

4. Contribution of the Six Food Groups

The relative contribution of each food group changed considerably between 2010 and 2023.

Food group 2010 share 2023 share Change in share

Fish	11.28%	13.27%	+1.99 pp
Meat	3.56%	3.44%	−0.12 pp
Milk	4.00%	11.68%	+7.68 pp
Eggs	0.92%	1.98%	+1.06 pp
Pulses	6.44%	9.51%	+3.07 pp
Cereals	73.80%	60.11%	−13.69 pp

Note: Shares are calculated relative to the combined protein supply of the six selected food groups. pp = percentage points.

Cereals accounted for **73.80%** of the protein supplied by the six selected food groups in 2010. By 2023, this share had fallen to **60.11%**, representing a decline of 13.69 percentage points. Thus, although cereals remained the dominant source, their relative contribution decreased substantially over the study period.

The largest increase in contribution was observed for **milk**, whose share increased from 4.00% in 2010 to 11.68% in 2023, an increase of 7.68 percentage points. Pulses also increased their contribution from 6.44% to 9.51%, while the share of fish increased from 11.28% to 13.27%.

The contribution of eggs increased from 0.92% to 1.98%. In contrast, the share of meat remained relatively stable, declining slightly from 3.56% to 3.44%.

Overall, the results indicate that the composition of protein supply became less concentrated in cereals over the study period, while milk, pulses, fish, and eggs accounted for increasing proportions of the protein supplied by the six selected groups.

5. Animal-Source and Plant-Source Protein

To further examine the composition of protein supply, the six food groups were divided into animal-source and plant-source categories.

Animal-source foods: Fish + Meat + Milk + Eggs

Plant-source foods: Pulses + Cereals

Source	2010	2023	Change
Animal-source	19.76%	30.38%	+10.62 pp
Plant-source	80.24%	69.62%	−10.62 pp

In 2010, approximately **19.76%** of the protein supplied by the six selected food groups came from animal-source foods, compared with **80.24%** from plant-source foods. By 2023, the animal-source share had increased to **30.38%**, while the plant-source share had declined to **69.62%**.

This represents a **10.62 percentage-point increase** in the contribution of animal-source foods and an equivalent decline in the contribution of plant-source foods.

The change was primarily associated with the increased contribution of milk, fish, and eggs. At the same time, the declining relative contribution of cereals reduced the overall share of plant-source protein, although cereals and pulses together still accounted for the majority of protein supplied by the six selected food groups in 2023.

6. Overall Findings

The analysis demonstrates a notable change in the composition of protein supply between 2010 and 2023. The most substantial increase in protein supply occurred in milk, followed by eggs, pulses, fish, and meat. Cereals were the only selected food group to experience a decline, although the reduction was relatively small in absolute terms.

Cereals remained the dominant source of protein throughout the period, but their contribution declined from 73.80% to 60.11% of the combined protein supply of the six selected groups. In contrast, the contribution of milk increased considerably, while pulses, fish, and eggs also gained relative importance.

The animal-source food share increased from 19.76% in 2010 to 30.38% in 2023, while the plant-source share decreased from 80.24% to 69.62%. Therefore, the results show a **shift in the composition of protein supply toward a more diversified combination of animal-source foods and pulses, alongside a reduced relative dependence on cereals.**

However, cereals continued to provide the largest share of protein among the six groups in 2023. The findings therefore indicate a change in the composition of protein supply rather than a complete replacement of cereal-based protein.

7. Conclusion

Between 2010 and 2023, Bangladesh experienced substantial changes in the composition of protein supplied by the six selected food groups. Milk showed the largest proportional increase, while eggs and pulses also experienced substantial growth. Fish increased steadily, whereas meat showed only a modest increase. Cereal-derived protein declined slightly, but cereals remained the largest contributor to protein supply.

The reduction in the cereal share, together with increases in milk, fish, eggs, and pulses, resulted in a greater contribution from animal-source foods and a lower relative contribution from plant-source foods. Nevertheless, plant-source foods continued to account for the majority of protein supplied by the six selected groups in 2023.

Overall, the analysis suggests that Bangladesh's protein supply became more diversified over the 2010–2023 period, with increasing contributions from several non-cereal food groups while cereals continued to play a dominant role.